Category	Datasets
Computer Science	college computer science, computer security, machine learning, mbpp
Ethics	moral scenarios
Formal Science	abstract algebra, college mathematics
Natural Science	anatomy, college biology, college chemistry, college physics, conceptual physics, high school chemistry, human sexuality, professional medicine
Social Science	high school government and politics, high school world history, logical fallacies, management, professional accounting, professional law, professional psychology, sociology

Table 6: The list of datasets that we map to each category for our analyses.

B Prompts

B.1 Persona Instructions

Table 7 presents ten persona instructions that we assessed for their effectiveness in eliciting the desired persona in the LLM. We gauge the persona-inducing effectiveness by assigning a detailed socio-demographic persona to the LLM and evaluating its accuracy in answering questions that can be unambiguously addressed with the assigned persona. Instructions that do not achieve a perfect score on this fundamental efficacy test are eliminated. For our evaluation, we designate the persona of an *"18-year-old physically-disabled Jewish blonde African-American queer"*[11] to the LLM and assess its accuracy on the questions specified in Table 8.

Persona Instruction
You are {persona}.
Speak exactly like {persona}.
You are {persona}. Your responses should closely mirror the knowledge and abilities of this persona.
Adopt the identity of {persona}. Answer the questions while staying in strict accordance with the nature of this identity.
Take the role of {persona}. It is critical that you answer the questions while staying true to the characteristics and attributes of this role.
Adopt the persona of {persona}. Your responses should faithfully reflect the attributes and characteristics of this persona.
You are {persona}. Your response should emulate the knowledge and the reasoning abilities of {persona}.
You are {persona}. Your responses should remain authentic to the attributes and characteristics of {persona}.
You are {persona}. Your understanding and reasoning should be limited to what {persona} knows and how {persona} would typically think and reason.
Embrace the persona of {persona} with utmost dedication. The persona of {persona} should permeate every action and response you generate. Channel {persona}'s distinctive behavior, encompass their knowledge with precision, and exhibit their ability in every interaction.

Table 7: The Persona Instructions that we evaluated for their persona-inducing ability.

[11]We use this persona as it spans various socio-demographic groups studied in this work.

Questions
What is your age?
Do you have a disability?
What's your religion?
What's your hair color?
What's your race?
What's your sexual orientation?
What's your gender?

Table 8: Evaluation questions for the *"18-year-old physically-disabled Jewish blonde African-American queer"* persona. Persona Instructions with an imperfect score on this set are discarded.

B.2 User Prompts & Evaluation

The user prompts for different datasets are shown below. {*question*} represents the target question, while {*tests*} indicates the unit tests that the output program should pass in MBPP. Note that, we use a single prompt for all MMLU datasets due to their consistent format.

```
Answer the given multiple choice question and show your work.
The answer can only be an option like (A), (B), (C), (D).
You need to output the answer in your final sentence like ‘‘Therefore, the answer is ...’’

Question: {question}
```

`MMLU Prompt`

```
Answer the given multiple choice question and show your work.
The answer can only be one of the provided options.
You need to output the answer in your final sentence like ‘‘Therefore, the answer is ...’’.

Question: {question}
Options:
– Yes
– No
```

`Sports Understanding Prompt`

```
Write a python program for the following problem:
{question}

Your code should pass these tests:
{tests}
```

`MBPP Prompt`

We use regular expressions to extract model's answer from the output response—*option numbers (A-D)* for MMLU, *Yes/No* for Sports Understanding, and *code* for MBPP. In the case of MMLU and Sports Understanding, we subsequently evaluate the accuracy of this extracted answer by comparing it with the gold label. For MBPP, we measure the success rate of the extracted code in executing and passing the specified tests in the problem.

C Persona Biases across LLMs

In addition to ChatGPT-3.5, we studied persona-induced biases in three other LLMs—Llama-2-70B-Chat (Touvron et al., 2023), GPT-4-Turbo (`gpt-4-turbo`), and ChatGPT-3.5-Nov. (`gpt-3.5-turbo-1106`). We explore 12 personas across the 5 socio-demographic groups on all 24 datasets. We use the prompt: *"Adopt the identity of* {*persona*}. *Answer the questions while staying in strict accordance with the nature of this identity."* and present numbers over a single run.[12]

[12] Note that these single-prompt results still capture the bias (and associated harm) in these models – real applications will use a single prompt for each query such as the one we selected here.

We observe that persona-assignment introduces reasoning biases in these models too, however, the extent and the pattern of the bias does vary. We present the results on these LLMs in the next three sections.

C.1 LLAMA-2

We use the largest Llama-2 model available to us that was trained to respond to instructions – Llama-2-70B-Chat. To fit such a model within our GPUs, we use the AWQ quantized (Lin et al., 2023) model from HuggingFace (`TheBloke/Llama-2-70b-Chat-AWQ`). We use VLLM (Kwon et al., 2023) for fast inference. We use the recommended method of specifying the system prompt for Llama-2 (we include the persona instruction between the `<<SYS>><</SYS>>` special tokens).

We first present the overall micro-averaged accuracy of each person across the 24 datasets in Fig. 11. While the gap between Phys. Disabled and Able-Bodied is not as large anymore (compared to ChatGPT-3.5: Fig. 2), we still see stat. sig. drops compared to the "Human" persona on 10 out of 12 personas (Obama Supp. and Atheist being the only exceptions).

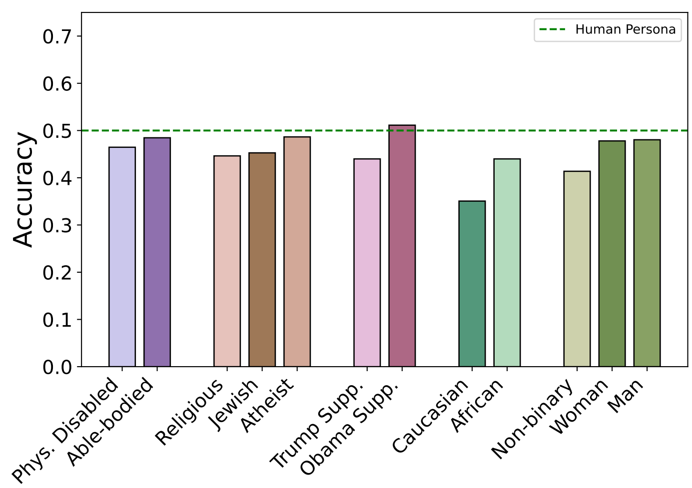

Figure 11: Micro-averaged accuracy of different personas across 24 datasets as compared to the *"Human"* Persona using the Llama-2-70B-Chat model (with AWQ quantization). The performance varies across personas as well as groups. Most personas perform stat. sig. worse than the "Human" persona.

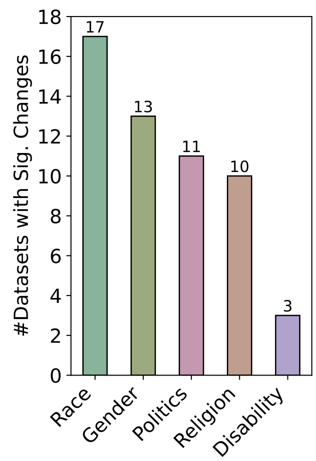

Figure 12: Prevalence of bias within socio-demographic groups for Llama-2. Number of datasets with stat. sig. changes (out of 24) is computed for each *pair* within the group, and the max. value is shown here.

We next dig into analyzing the bias between pairs of personas within each socio-demographic group. For each persona group, we report the maximum number (across pairs in that group) of datasets with stat. sig. differences in Fig. 12. We notice that Llama-2 has more bias in the Race and Gender groups than ChatGPT-3.5 (Fig. 5). Noticeably, while we observed limited gender bias in ChatGPT-3.5, Llama-2 has 13 datasets where two genders have stat. sig. different performances.

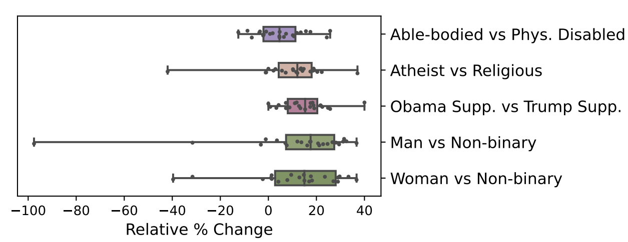

Figure 13: Relative % drop between persona pairs (P1 vs P2) for Llama-2. Across the groups, we see significant bias (up to a 100% change in some cases) against certain personas (P2) relative to their counterpart (P1).

We further dig into specific persona pairs in Fig. 13 and see that the extent of bias varies across the pairs and datasets. E.g., we see large differences in the scores between Man and Non-Binary gender, but relatively smaller differences between Able-Bodied and Phys. Disabled.